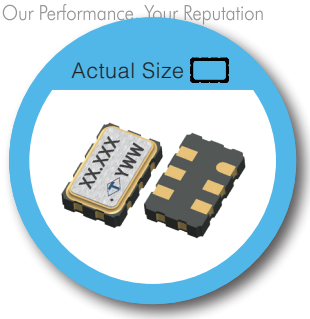
TAITIEN
Our Performance, Your Reputation

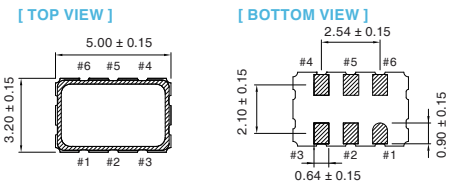
OW-M Type
5.0 x 3.2 mm SMD CMOS Crystal Oscillator



FEATURE
- Industry Standard 5.0 x 3.2 Hermitically sealed ceramic package
- Very low phase jitter: < 1pS(0.6pS, typ.)RMS
- Any frequency between 10MHz and 250MHz
- Fast delivery

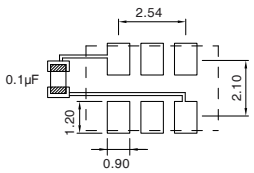
TYPICAL APPLICATION
- High-Speed Gigabit Ethernet, Fiber Channel, Storage
- Enterprise Server, SAS/SATA
- Microprocessors/DSP/FPGA
- Broadband Access
- Smart Grid

DIMENSION (mm)



Pin#	Function
1	NC/Tri-State
2	Tri-State/NC
3	GND
4	Output
5	Comp. Output
6	VDD

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1μF as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter	CMOS 3.3V Min.	CMOS 3.3V Max.	CMOS 2.5V Min.	CMOS 2.5V Max.	Unit
Supply Voltage Variation (VDD)	VDD-5%	VDD+5%	VDD-5%	VDD+5%	V
Frequency Range	10	250	10	250	MHz
Standard Frequency	106.25, 125, 133.33, 150, 155.52, 158.25, 187.5, 212.5				MHz
Supply Current 10MHz≦ Fo≦ 250MHz	-	30	-	30	mA
Output Level Output High (Logic "1")	2.97	-	2.25	-	V
Output Low (Logic "0")	-	0.33	-	0.25	V
Transition Time : Rise/ Fall Time +	-	3	-	3	nSec
Start Time	-	10	-	10	mSec
Tri-State(Input to Pin 2 or Pin 1)					
Enable (High voltage or floating)	2.31	-	1.75	-	V
Disable (Low voltage or GND)	-	0.99	-	0.75	V
RMS Phase Jitter (Integrated 12 kHz ~ 20 MHz) (At Integer Mode)	-	1.0	-	1.0	pSec
Phase Noise @156.25 MHz 100 Hz	-	-75	-	-75	dBc/Hz
1 kHz	-	-105	-	-105	dBc/Hz
10 kHz	-	-120	-	-120	dBc/Hz
Aging (@25℃ 1st year)	-	± 3	-	± 3	ppm
Storage Temp. Range	-55	125	-55	125	℃

+Transition times are measured between 20% and 80% of VDD

XO-0116
Rev(10) 05/2024

FREQ. STABILITY vs. TEMP. RANGE

Temp. (℃) \ ppm	±25	±50
-10 ~ +60	○	○
-20 ~ +70	○	○
-40 ~ +85	△	○
-40 ~ +105	X	△

* ○: Available △:Conditional X: Not available
* Inclusive of calibration @ 25 °C, operating temperature range, input voltage variation, load variation, aging (1st year), shock, and vibration

Note: not all combination of options are available. Other specifications may be available upon request.